

Lesson 17: Central Limit Theorem

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Learning Objectives

1. Calculate probability of a sample mean using a population mean and variance with unknown distribution
2. Use the Central Limit Theorem to construct the Normal approximation of the Binomial and Poisson distributions

The Central Limit Theorem

Theorem 1: Central Limit Theorem (CLT)

Let X_i be iid rv's with common mean μ and variance σ^2 , for $i = 1, 2, \dots, n$. Then

$\left\{ \begin{array}{l} \text{independent \&} \\ \text{identically} \\ \text{distributed} \end{array} \right\}$

$$\sum_{i=1}^n X_i \rightarrow N(n\mu, n\sigma^2)$$

→ converges in distribution
(\& probability) as $n \rightarrow \infty$

- X_i ' do NOT need to be normally distributed
- don't need a known distribution of X
- in application: when n is "large", we can use Normal approx.

Extension of the CLT

Corollary 1

Let X_i be iid rv's with common mean μ and variance σ^2 , for $i = 1, 2, \dots, n$. Then

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n} \rightarrow N\left(\mu, \frac{\sigma^2}{n}\right)$$

★ sampling distribution of the sample means ★

BSTA 511

used in 512/513

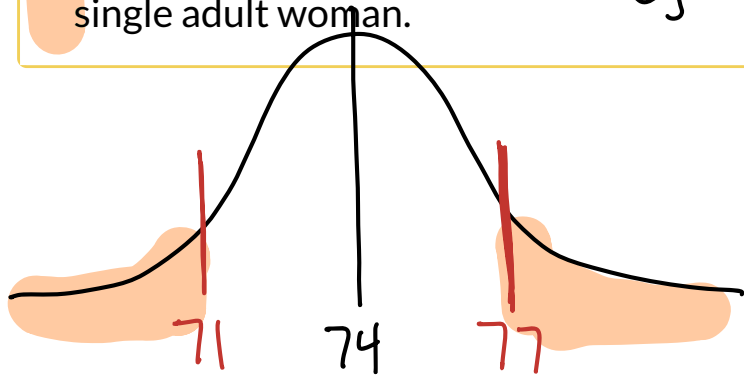
Example of Corollary in use

Example 1

$\bar{X}$

According to a large US study, the mean resting heart rate of adult women is about 74 beats per minutes (bpm), with standard deviation 13 bpm (NHANES 2003-2004).

1. Find the probability that the average resting heart rate for a random sample of 36 adult women is more than 3 bpm away from the mean.
2. Repeat the previous question for a single adult woman.



$n \geq 30$ b/c 36 women is sample

$$\mu = 74 \quad \sigma = 13 \quad Z^* = \mu \pm 3 \\ = 71 \text{ or } 77$$

$$\textcircled{1} P(\bar{X} < 71 \text{ or } \bar{X} > 77)$$

$$\bar{X} \sim \text{Normal}(\mu = 74, sd = \frac{\sigma}{\sqrt{n}} = \frac{13}{\sqrt{36}})$$

$$\rightarrow P(\bar{X} < 71) + P(\bar{X} > 77)$$

$$= 2 \cdot P(\bar{X} < 71)$$

$$= 2 \cdot \text{pnorm}(x = 71, \text{mean} = 74, \\ sd = 13 / \text{sqrt}(36))$$

$$= 0.164$$

$$\textcircled{2} n = 1 < 30, \text{ no normal approx}$$

Rule of thumb: $n \geq 30$
to use CLT

Example of CLT for exponential distribution

Example 2

Let $X_i \sim \text{Exp}(\lambda)$ be iid RVs for $i = 1, 2, \dots, \underline{n}$. Then

$$\underline{\sum_{i=1}^n X_i} \rightarrow$$

$$\mu = \frac{1}{\lambda}$$
$$\sigma^2 = \frac{1}{\lambda^2}$$

$$\sum_{i=1}^n X_i \longrightarrow \text{Normal}(\underline{n\mu}, \underline{n\sigma^2})$$

↓
converges in
distribution as
 $n \rightarrow \infty$

$$n\mu = \frac{n}{\lambda} \quad n\sigma^2 = \frac{n}{\lambda^2}$$

$$\sum_{i=1}^n X_i \rightarrow N\left(\frac{n}{\lambda}, \frac{n}{\lambda^2}\right)$$

CLT for Discrete RVs

1. **Binomial rv's**: Let $X \sim \text{Bin}(n, p)$

- $X = \sum_{i=1}^n X_i$, where X_i are iid Bernoulli(p)
- Rule of thumb: $np \geq 10$ and $n(1-p) \geq 10$ to use Normal approximation

$\sum_{i=1}^n X_i \rightarrow N(np, np(1-p))$

sum of Bern is a Binomial

$\hat{p} = \frac{\sum_{i=1}^n X_i}{n} \rightarrow N\left(p, \frac{p(1-p)}{n}\right)$

2. **Poisson rv's**: Let $X \sim \text{Poisson}(\lambda)$

- $X = \sum_{i=1}^n X_i$, where X_i are iid Poiss(1)
- Recall from ~~Chapter 18~~ ^{Lesson 15} that if $X_i \sim \text{Poiss}(\lambda_i)$ and X_i independent, then $\sum_{i=1}^n X_i \sim \text{Poiss}(\sum_{i=1}^n \lambda_i)$
- Rule of thumb: $\lambda \geq 10$ to use Normal approximation

$\sum_{i=1}^n X_i \rightarrow N(\lambda, \lambda^2)$

mean var

Larger example (1/7)

Example 3

Suppose that the probability of developing a specific type of breast cancer in women aged 40-49 is 0.001. Assume the occurrences of cancer are independent. Suppose you have data from a random sample of 20,000 women aged 40-49.

1. How many of the 20,000 women would you expect to develop this type of breast cancer, and what is the standard deviation?
2. Find the **exact** probability that more than 15 of the 20,000 women will develop this type of breast cancer.
3. Use the CLT to find the **approximate** probability that more than 15 of the 20,000 women will develop this type of breast cancer.
4. Use the CLT to approximate the following probabilities, where X is the number of women that will develop this type of breast cancer.
 - a. $\mathbb{P}(15 \leq X \leq 22)$
 - b. $\mathbb{P}(X > 20)$
 - c. $\mathbb{P}(X < 20)$
5. Find the **approximate** probability that more than 15 of the 20,000 women will develop this type of breast cancer - not using the CLT!
6. Use the CLT to approximate the approximate probability in the previous question!

Larger example (2/7)

Example 3

Suppose that the probability of developing a specific type of breast cancer in women aged 40-49 is 0.001. Assume the occurrences of cancer are independent. Suppose you have data from a random sample of 20,000 women aged 40-49.

1. How many of the 20,000 women would you expect to develop this type of breast cancer, and what is the standard deviation?

Let $X = \underline{\text{\# women that develop type of breast cancer}}$

$$X \sim \text{Bin}(n=20,000, p=0.001)$$

$$\mu = np = 20000 \cdot 0.001 = 20$$

$$\begin{aligned}\sigma &= \sqrt{np(1-p)} = \sqrt{20000 \cdot 0.001(1-0.001)} \\ &= 4.4699\end{aligned}$$

Larger example (3/7)

Example 3

Suppose that the probability of developing a specific type of breast cancer in women aged 40-49 is 0.001. Assume the occurrences of cancer are independent. Suppose you have data from a random sample of 20,000 women aged 40-49.

2. Find the **exact** probability that more than 15 of the 20,000 women will develop this type of breast cancer.

$$P(X > 15) = \sum_{x=16}^{20000} \binom{20000}{x} 0.001^x 0.999^{20000-x}$$

or ≥ 16

$$= \text{pbinom}(q = 15, n = 20000, p = 0.001, \text{lower.tail} = F)$$

$$= 0.843616$$

lower.tail = F
 $P(X > 15)$

true:
 $P(X \leq 15)$

Larger example (4/7)

$$P(\underline{X} \leq \underline{k}) = P(X \leq k + 0.5)$$

$$P(X \geq \underline{k}) = \underline{P(X \geq k - 0.5)}$$



Example 3

Suppose that the probability of developing a specific type of breast cancer in women aged 40-49 is 0.001. Assume the occurrences of cancer are independent. Suppose you have data from a random sample of 20,000 women aged 40-49.

3. Use the CLT to find the approximate probability that more than 15 of the 20,000 women will develop this type of breast cancer.

$$X \rightarrow N(\mu = np, \sigma^2 = np(1-p))$$
$$N(20, 19.98)$$

★ check

$$np \geq 10 \quad np(1-p) \geq 10$$

$$20 \geq 10 \quad 19.98 \geq 10$$

so we can use CLT

$$P(X > 15) = P(X \geq \underline{15.5})$$

or $\underline{P(X \geq 16)}$

$$= \text{pnorm}(q = 15.5, \text{mean} = 20, \text{sd} = 4.4699, \text{lower.tail} = F)$$

$$= 0.8438$$

Larger example (5/7)

Example 3

Suppose that the probability of developing a specific type of breast cancer in women aged 40-49 is 0.001. Assume the occurrences of cancer are independent. Suppose you have data from a random sample of 20,000 women aged 40-49.

4. Use the CLT to approximate the following probabilities, where X is the number of women that will develop this type of breast cancer.

a. $\mathbb{P}(15 \leq X \leq 22)$

b. $\mathbb{P}(X > 20)$

c. $\mathbb{P}(X < 20)$



Larger example (6/7)

Example 3

Suppose that the probability of developing a specific type of breast cancer in women aged 40-49 is 0.001. Assume the occurrences of cancer are independent. Suppose you have data from a random sample of 20,000 women aged 40-49.

5. Find the **approximate** probability that more than 15 of the 20,000 women will develop this type of breast cancer - not using the CLT!

Poisson approx of Binomial

Larger example (7/7)

Example 3

Suppose that the probability of developing a specific type of breast cancer in women aged 40-49 is 0.001. Assume the occurrences of cancer are independent. Suppose you have data from a random sample of 20,000 women aged 40-49.

6. Use the CLT to approximate the approximate probability in the previous question!